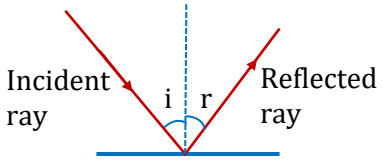
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Plane Mirror

1. Law of reflection states that when a light ray is incident on a reflecting surface then angle of incidence (i) = Angle of reflection (r). If incident ray is given by $\left[\frac{\hat{i}}{2}+\frac{\sqrt{3}}{2}\hat{j}\right]$ and reflected ray is given by $\left[\frac{\hat{i}}{2}-\frac{\sqrt{3}}{2}\hat{j}\right]$, then angle of incidence is :-

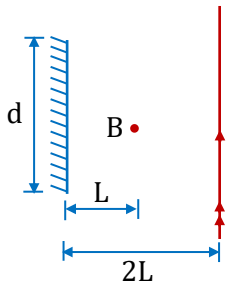


(A) 30° (B) 45° (C) 60° (D) 120°

PGO115

2. A point source of light B is placed at a distance L in front of the centre of a mirror of width d hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance $2L$ from it as shown. The greatest distance over which he can see the image of the light source in the mirror is :-

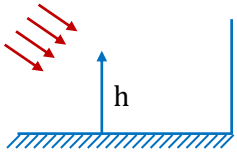
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(A) $d/2$ (B) d (C) $2d$ (D) $3d$

PGO117

3. Sun rays are reflected from a horizontal mirror and fall on a vertical screen. An object is placed perpendicular to mirror at large distance from the screen. If object is of length 'h' then length of the shadow on the screen of image will be

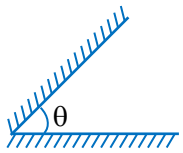


(A) h (B) $2h$ (C) $3h$ (D) ∞

PGO122

For More Join: @IITJEE_Advanced

4. Two mirrors are inclined at an angle θ as shown in the figure. Light ray is incident parallel to one of the mirrors. The ray will start retracing its path after third reflection if :-



(A) θ = 45° (B) θ = 30° (C) θ = 60° (D) all three

PGO124

5. If two mirrors are kept at 60° to each other, then the number of images formed by them is :-

(A) 5 (B) 6 (C) 7 (D) 8

PGO127

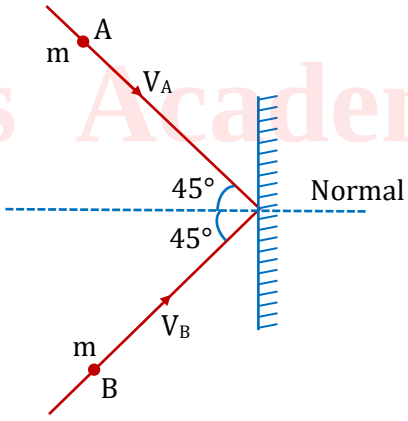
6. A point object approaches a plane mirror with a speed of 10 ms^{-1}, while the image recedes away from the mirror with a speed of 6 ms^{-1} (as seen by stationary observer). The direction and magnitude of the velocity of mirror is :-

(A) towards the object, 8 ms^{-1} (B) towards the image, 6 ms^{-1}

(C) away from the object, 8 ms^{-1} (D) away from the object, 2 ms^{-1}

PGO128

7. Two bodies *A* and *B* are moving towards a plane mirror with speeds v_A and v_B respectively as shown in the figure. The speed of image of *A* with respect to the body *B* is :-



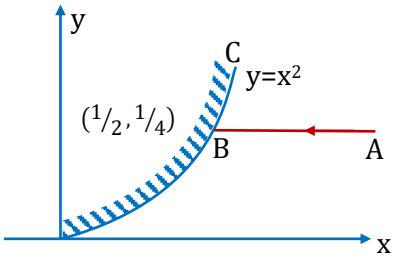
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(A) $v_A + v_B$ (B) $|\vec{v}_A + \vec{v}_B|$ (C) $\sqrt{v_A^2 + v_B^2}$ (D) $\sqrt{v_A^2 - v_B^2}$

PGO131

Spherical Mirror

8. In the given figure, incident ray *AB* is parallel to *x*-axis. The angle of reflection is :-

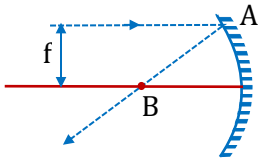


(A) 30° (B) 60° (C) 45° (D) None of these

PGO137

For More Join: @IITJEE_Advanced

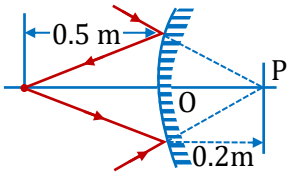
9. A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point *B* to the distance of the focus from the centre of curvature is (*AB* is the reflected ray) :-



(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$

PGO138

10. A beam of convergent light converges to a point 0.5 *m* in front of the mirror after reflection at a convex mirror but in the absence of the mirror the beam converges to a point 0.2 *m* behind the mirror. The radius of curvature of the mirror is :-



(A) 20 *cm* (B) 50 *cm* (C) 66.67 *cm* (D) 28.57 *cm*

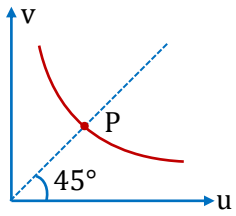
PGO144

11. A convex mirror has a radius of curvature of 20 *cm*. An object is placed in front at such a distance from the mirror that the size of the image is exactly half that of the object. The object distance must be :

(A) 20 *cm* (B) 30 *cm* (C) 10 *cm* (D) 40 *cm*

PGO146

12. The graph shows the variation of *v* with change in *u* for a mirror. Points plotted above the point *P* on the curve are for values of *v*



(A) smaller than *f* (B) smaller than 2*f* (C) larger than 2*f* (D) larger than *f*

PGO147

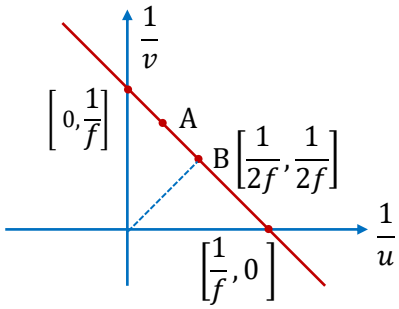
13. An infinitely long rod lies along the axis of a concave mirror of focal length *f*. The near end of the rod is at distance $u > f$ from the mirror. Its image will have a length–

(A) $\frac{uf}{u-f}$ (B) $\frac{uf}{u+f}$ (C) $\frac{f^2}{u+f}$ (D) $\frac{f^2}{u-f}$

PGO148

For More Join: @IITJEE_Advanced

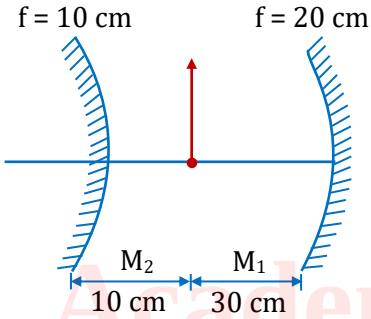
14. Figure shows $\frac{1}{v}$ vs $\frac{1}{u}$ curve for convex mirror. Nature of image at point A is :-



(A) Real, erect, magnified
(B) virtual, inverted, magnified
(C) virtual, inverted, diminished
(D) virtual, erect, diminished

PGO149

15. In figure, find the total magnification after two successive reflections first on M_1 and then on M_2 :-



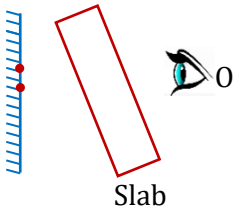
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(A) +1 (B) –2 (C) +2 (D) –1

PGO155

Refraction from Plane Surface

16. The observer at O views two closely spaced spots on a vertical wall through an angled glass slab as shown. As seen by observer, the spots appear.



(A) shifted upward
(B) shifted downward
(C) spaced farther apart
(D) spaced closer together

PGO161

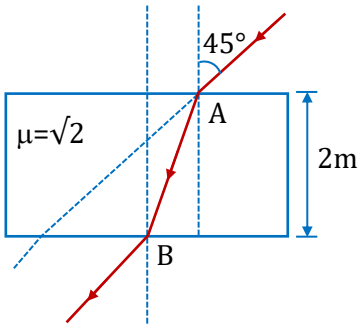
17. A light ray is travelling from air to a liquid of refractive index $\sqrt{3}$. If the incident beam is rotated at a rate of 3 rad/sec then the angular speed of refracted beam when angle of incidence is 60° is :-

(A) 3 rad/sec (B) $\sqrt{3}$ rad/sec (C) 1 rad/sec (D) rad/sec

PGO164

For More Join: @IITJEE_Advanced

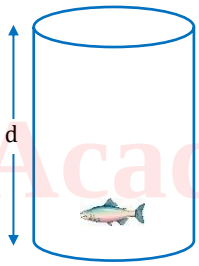
18. Find the time taken by a monochromatic light to cross a transparent slab of width $2m$ and *R.I.* = $\sqrt{2}$ as shown in figure :-



(A) $\frac{2\sqrt{2}}{3}\times10^{-8}s$ (B) $\frac{4\sqrt{2}}{3}\times10^{-8}s$ (C) $\frac{4}{3\sqrt{3}}\times10^{-8}s$ (D) $\frac{4\sqrt{2}}{3\sqrt{3}}\times10^{-8}s$

PGO165

19. A fish floats in liquid with its eye at the centre of an opaque walled full tank of liquid of circular cross section. When the fish look upwards, it can see a fish-eye view of the surrounding scene i.e. it is able to view the entire space above the liquid surface. The diameter of the tank is 30 *cm*, and the critical angle for liquid is 37°. At what maximum depth below the surface of the liquid, *d*, must the fish be floating? $\left(\sin 37^\circ = \frac{3}{5}\right)$



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(A) 16 *cm* (B) 20 *cm* (C) 11.25 *cm* (D) 25 *cm*

PGO171

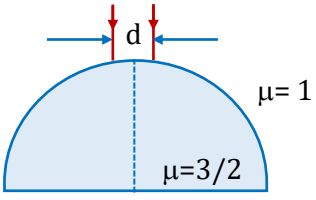
Refraction through Curved Surface

20. A light ray is incident on a transparent sphere of index =$\sqrt{2}$, at an angle of incidence = 45°. What is the deviation of a tiny fraction of the ray, which enters the sphere, undergoes two internal reflections and then refracts out into air ?

(A) 270° (B) 240° (C) 120° (D) 180°

PGO174

21. A beam of diameter '*d*' is incident on a glass hemisphere as shown. If the radius of curvature of the hemisphere is very large in comparison to *d*, then the diameter of the beam at the base of the hemisphere will be :-

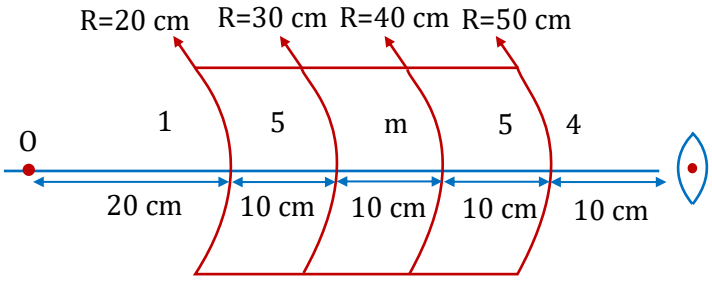


(A) $\frac{3}{4}d$ (B) d (C) $\frac{d}{3}$ (D) $\frac{2}{3}d$

PGO189

For More Join: @IITJEE_Advanced

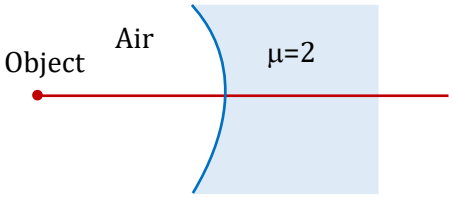
22. In the arrangement shown, separation between observer and object as seen by observer is



(A) 60 *cm* (B) more than 60 *cm* (C) less than 60 *cm* (D) depends on μ

PGO192

23. A concave spherical surface with radius *R* separates a medium with index of refraction 2 from air. If an object is moved toward the surface from far away along the central axis, its image :



(A) Changes from virtual to real when it is $R/2$ from the surface

(B) Changes from virtual to real when it is $2R$ from the surface

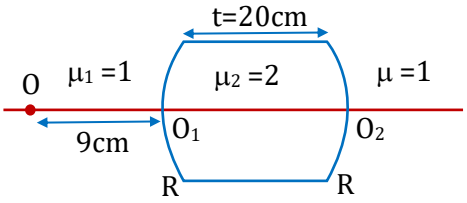
(C) Changes from real to virtual when it is $R/2$ from the surface

(D) Remains virtual

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PGO193

24. In the given figure, if object is placed at a distance 9 *cm* from O_1 then find the image distance (in *cm*) from O_2. (R = 9 *cm* for both surface)



(A) 9 *cm* (B) 20 *cm* (C) 18 *cm* (D) infinity

PGO196

Lens

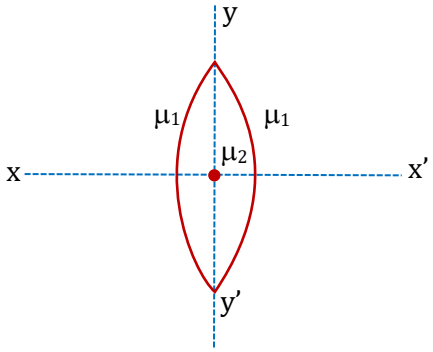
25. A convergent (biconvex) lens is placed inside a jar filled with a liquid. The lens has focal length 20*cm*, when in air and its material has a refractive index of 1.5. If the liquid has a refractive index of 1.6, the focal length of the lens while in the jar, is :-

(A) –110 *cm* (B) –130 *cm* (C) –160 *cm* (D) –180 *cm*

PGO191

For More Join: @IITJEE_Advanced

26. An equi-convex lens of refractive index μ_2 and focal length f (in air) is kept in medium of refractive index μ_1.



	Column-A		**Column-B**
(P)	If lens is cut in two equal parts by a plane yy'. ($\mu_1 = 1$)	(1)	Lens will act as a glass slab
(Q)	If lens is cut in two equal parts by a plane xx'. ($\mu_1 = 1$)	(2)	Lens will be converging and focal length will change.
(R)	If $\mu_1 = \mu_2$.	(3)	Lens will be converging and focal length will remain same
(S)	If $\mu_1 > \mu_2$.	(4)	Lens will be diverging and focal length will change.

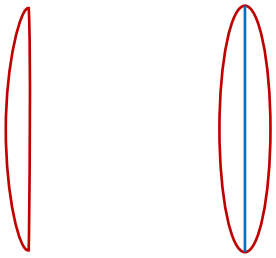
Codes :

	P	**Q**	**R**	**S**
(A)	2	4	1	3
(B)	2	3	1	4
(C)	3	4	2	1
(D)	1	2	3	4

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PGO200

27. You are given two thin identical plano-convex lenses, one of which is shown in the figure. When you place an object 20 *cm* to the left of a single plano-convex lens, the image appears 40 *cm* to the right of the lens. You then arrange the two plano-convex lenses back to back to form a double convex lens. If the object is at 20 *cm* to the left of this new lens, what is the approximate location of the image?



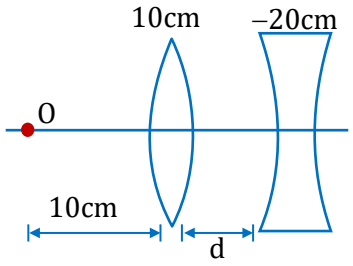
Plano-convex **Double convex**

(A) 6.7 *cm* to the right of the lens.
(B) 10 *cm* to the right of the lens.
(C) 20 *cm* to the right of the lens.
(D) 80 *cm* to the right of the lens.

PGO206

For More Join: @IITJEE_Advanced

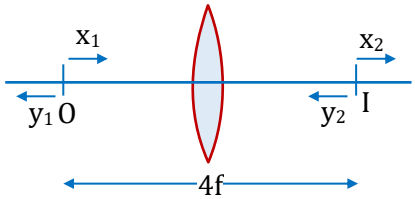
28. What should be the value of distance *d* so that final image is formed on the object itself ? (Focal lengths of the lenses are written on the lenses).



(A) 10 *cm* (B) 20 *cm* (C) 5 *cm* (D) none of these

PGO207

29. In a converging lens of focal length *f* and the distance between real object and its real image is 4*f*. If the object moves x_1 distance towards lens its image moves x_2 distance away from the lens and when object moves y_1 distance away from the lens its image moves y_2 distance towards the lens, then choose the correct option :-



(A) $x_1 > x_2$ and $y_1 > y_2$ (B) $x_1 < x_2$ and $y_1 < y_2$
(C) $x_1 < x_2$ and $y_1 > y_2$ (D) $x_1 > x_2$ and $y_2 > y_1$

PGO209

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30. In the displacement method, a convex lens is placed in between an object and a screen. If the magnification in the two positions are m_1 and $m_2, (m_1 > m_2)$, the distance between the two positions of the lens is *x*, the focal length of the lens is

(A) $\frac{x}{m_1 + m_2}$ (B) $\frac{x}{m_1 - m_2}$ (C) $\frac{x}{(m_1 + m_2)^2}$ (D) $\frac{x}{(m_1 - m_2)^2}$

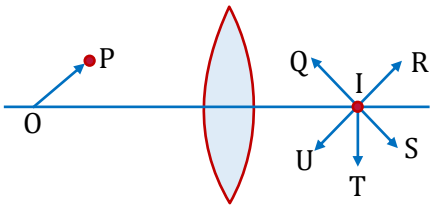
PGO213

31. The focal length of a convex lens is *f* and the distance of an object from the principal focus is *x*. The ratio of the size of the real image to the size of the object is

(A) f / x (B) x / f (C) $\frac{f + x}{f}$ (D) $\frac{f}{f + x}$

PGO214

32. A point object *O* moves from the principal axis of a converging lens in a direction *OP*. *I* the image of *O*, will move initially in the direction :



(A) *IQ* (B) *IR* (C) *IS* (D) *IU*

PGO215

For More Join: @IITJEE_Advanced

33. When the object is at distances u_1 and u_2 the images formed by the same lens are real and virtual respectively and of the same size. Then focal length of the lens is :

(A) $\frac{1}{2}\sqrt{u_1u_2}$ (B) $\frac{u_1+u_2}{2}$ (C) $\sqrt{u_1u_2}$ (D) $\sqrt{(u_1+u_2)}$

PGO216

Prism

34. A ray of light is incident at 60° on a prism of refracting angle 30°. The emerging ray is at an angle 30° with the incident ray. The value of refractive index of the prism is :-

(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\frac{2}{\sqrt{3}}$

PGO178

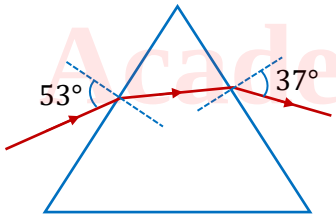
35. An isosceles glass prism having refractive index μ has one of its faces coated with silver. A ray of light is incident normally on the other face (which is equal to the silvered face). The ray of light is reflected twice on the same sized faces and then emerges through the base of the prism perpendicularly. The angles of prism are :-

(A) 40°, 70°, 70° (B) 50°, 65°, 65° (C) 36°, 72°, 72° (D) data insufficient

PGO181

36. A ray incident at an angle 53° on a prism emerges at an angle 37° as shown. If the angle of incidence is made 50°, which of the following is a possible value of the angle of emergence?

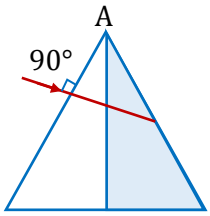
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(A) 35° (B) 42° (C) 40° (D) 38°

PGO182

37. A ray of light is incident normally on the first refracting face of the prism of refracting angle A. The ray of light comes out at grazing emergence. If one half of the prism (shaded position) is knocked off, the same ray will :–



(A) Emerge at an angle of emergence $\sin^{-1}\left(\frac{1}{2}\sec A/2\right)$

(B) Not emerge out of the prism

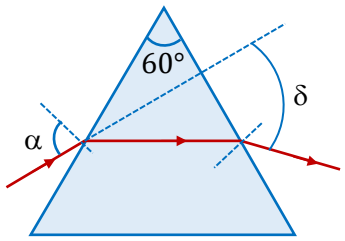
(C) Emerge at an angle of emergence $\sin^{-1}\left(\frac{1}{2}\sec A/4\right)$

(D) None of these

PGO183

For More Join: @IITJEE_Advanced

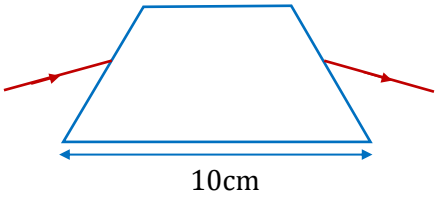
38. A light ray is incident on a triangular prism of refracting angle 60° at an angle of incidence 37° as shown in figure. The refractive index of prism material is 1.5 and surrounding medium is having refractive index 1. The angle δ through which the light ray deviates from its path is :
[(sin 37° = 0.6, sin (23.6°) = 0.4, sin (36.4°) = 0.59, $\sin^{-1}$ (0.885) = 62.3°]



(A) 39.3° (B) 52° (C) 42.4° (D) None of these

PGO185

39. Consider an isosceles trapezoid of material of refractive index √2 and dimensions of sides being 5 *cm*, 5 *cm*, 10 *cm* and 5 *cm*. The angle of minimum deviation by this arrangement when light is incident from air and emerges in air is

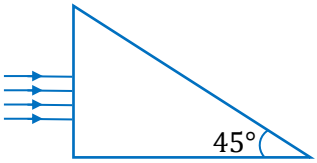


(A) 22.5° (B) 30° (C) 45° (D) 60°

PGO186

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40. A beam of light consisting of red, green and blue and is incident on a right-angle prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. The prism will



(A) separate part of the red color from the green and blue colors
(B) separate part of the blue color from the red and green colors
(C) separate all the three colors from the other two colors
(D) not separate even partially, any colors from the other two colors

PGO220

41. Chromatic aberration will be absent if for two thin lenses in contact :-
(A) $(\omega_1/F_1) + (\omega_2/F_2) = 0$ (B) $(\omega_1/F_2) + (\omega_2/F_1) = 0$
(C) $(F_1/\omega_2) + (F_2/\omega_1) = 0$ (D) $(\omega_1/\omega_2) + (F_1 + F_2) = 0$

PGO221

42. An achromatic convergent doublet of two lenses in contact has a power of + 2*D*. The convex lens has a power + 5*D*. What is the ratio of the dispersive powers of the convergent and divergent lenses
(A) 2 : 5 (B) 3 : 5 (C) 5 : 2 (D) 5 : 3

PGO224

For More Join: @IITJEE_Advanced

43. It is desired to make an achromatic combination of two lenses (L_1 and L_2) made of materials having dispersive powers ω_1 and ω_2 ($<\omega_1$). If the combination of lenses is converging then

(A) L_1 is converging

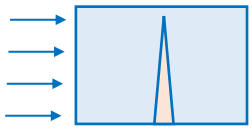
(B) L_2 is converging

(C) Power of L_1 is greater than the power of L_2

(D) None of these

PGO225

44. A thin isosceles prism with angle 4° and refractive index 5/4 is placed inside a transparent tube with liquid (refractive index = 1.5) as shown. The deviation of light due to prism will be :-



(A) 0.8° upward (B) 0.8° downward (C) 0.67° upward (D) 0.67° downward

PGO226

Defects of Eye and Optical Instruments

45. A long-sighted person has a minimum distance of distinct vision of 50 *cm*. He wants to reduce it to 25 *cm*. He should use a

(A) Concave lens of focal length 50 *cm* (B) Convex lens of focal length 25 *cm*

(C) Convex lens of focal length 50 *cm* (D) Concave lens of focal length 25 *cm*

PGO227

Rankers Academy JEE

46. The focal length of the objective of a compound microscope is f_o and its distance from the eyepiece is L. The object is placed at a distance u from the objective. For proper working of the instrument,

(A) $L < u$ (B) $L > u$ (C) $f_o < L < 2f_o$ (D) $L > 2f_o$

PGO231

47. A compound microscope has a magnification of 30. The focal length of the eyepiece is 5 *cm*. If the final image is formed at the least distance of vision (25 *cm*), the magnification produced by the objective is

(A) 5 (B) 7.5 (C) 10 (D) 15

PGO232

48. A person can see objects only at a distance greater than 40 *cm*. He is advised to use lens of power

(A) + 2.5 *D* (B) + 1.5 *D* (C) – 2.5 *D* (D) – 6.25 *D*

PGO233

49. A person cannot see clearly objects at a distance less than 100 *cm*. The power of the spectacles required to see clearly objects at 25 *cm* is

(A) + 1 *D* (B) + 3 *D* (C) + 4 *D* (D) + 2 *D*

PGO234

50. When the length of an astronomical telescope tube increases its magnifying power

(A) Decreases (B) Increases

(C) Does not change (D) May increase or decrease

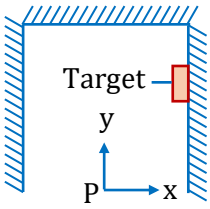
PGO237

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

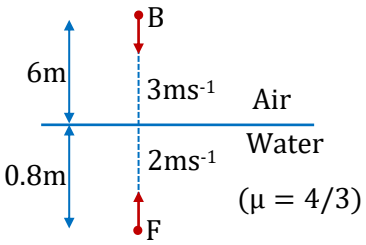
1. The drawing shows a top view of a square room. One wall is missing and the other three are each mirrors. From point *P* in the center of the open side, a laser is fired, with the intent of hitting a small target located at the center of one wall. Identify vector in whose direction the laser can be fired and score a hit, assuming that the light does not strike any mirror more than one.



(A) $\frac{\hat{i}}{3}+\hat{j}$ (B) $\hat{i}+\hat{j}$ (C) $-\hat{i}+\hat{j}$ (D) $-\hat{i}+\frac{\hat{j}}{3}$

PGO118

2. A fish, *F* in the pond, is at a depth of 0.8 *m* from water surface and is moving vertically upwards with velocity 2 ms^{-1}. At the same instant, a bird *B* is at a height of 6 *m* from water surface and is moving downwards with velocity 3 ms^{-1}. At this instant both are on the same vertical lines as shown in the figure. Which of the following statement(s) is(are) correct?



(A) Height of *B*, observed by *F* (from itself) is equal to 8.00 *m*.

(B) Depth of *F*, observed by *B* (from itself) is equal to 6.60 *m*.

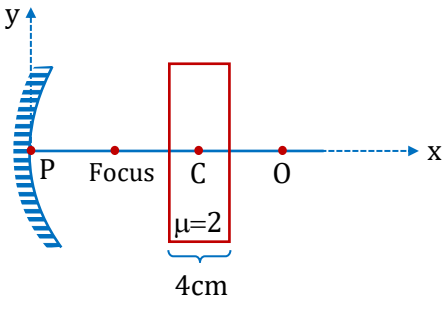
(C) Velocity of *B*, observed by *F* (relative to itself) is equal to 5.00 ms^{-1}.

(D) Velocity of *F*, observed by *B* (relative to itself) if equal to 4.50 ms^{-1}.

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PGO160

3. A concave mirror of focal length 50 *cm* is placed at origin as shown in diagram. A point sized object is placed at *O* (x = 152 *cm*). A glass slab of thickness 4 *cm*, μ = 2, is placed symmetrically around *C*. Light can not reach from object to mirror without passing through slab. Choose the **CORRECT** option(s) :



(A) Image made by concave mirror will be at *x* = 75 *cm*

(B) Final image made by entire optical system will be at *x* = 77 *cm*.

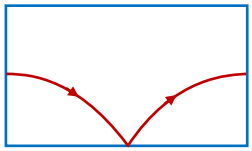
(C) Final image of the system will be real.

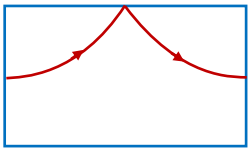
(D) Final image of the system will be virtual.

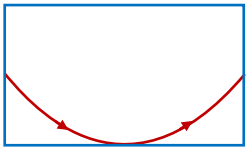
PGO168

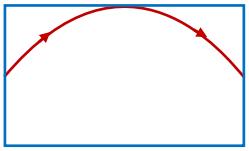
For More Join: @IITJEE_Advanced

4. A cubic container is filled with a liquid whose refractive index increases linearly from top to bottom. Which of the following figures may represent the path of a ray of light inside the liquid?

(A) 

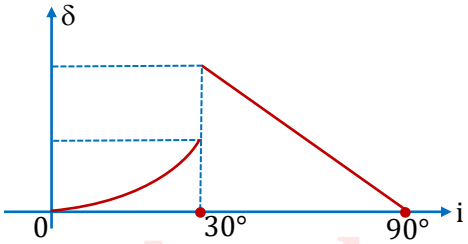
(B) 

(C) 

(D) 

PGO172

5. Figure shows graph of angle of deviation *v/s* angle of incidence for a light ray. Incident ray goes from medium 1 (μ_1) to medium 2 (μ_2). Mark the **correct** option(s).



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(A) $\frac{\mu_1}{\mu_2} = \frac{1}{2}$

(B) Critical angle is 30°

(C) $\mu_1 > \mu_2$

(D) Maximum deviation is 120°

PGO173

6. In the figure (*a*) light is incident at an angle θ which is slightly greater than the critical angle. Now, keeping the incident fixed a slab of refractive index n_3 ($n_3 < n_2$) is placed on surface *AB* in fig. (*b*). Which of the following statements are correct :

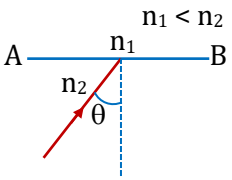


Fig. (a)

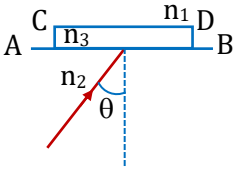


Fig. (b)

(A) total internal reflection occurs at *AB* for $n_3 < n_1$

(B) total internal reflection occurs at *AB* for $n_3 > n_1$

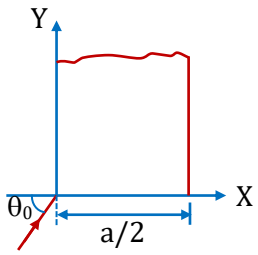
(C) the ray will return back to the same medium for all values of n_3

(D) total internal reflection occurs at *CD* for $n_3 < n_1$

PGO176

For More Join: @IITJEE_Advanced

7. A glass slab is of thickness $d = a/2$ is shown in figure. The refractive index of the slab varies according to relation $\mu = \frac{a}{a-x}$ where a is positive constant. A ray of light enters the slab at origin at an angle $\theta_0 \to \pi/2$, then choose the **CORRECT** statement (s):-



(A) The path of the light ray inside medium is a circle given by $(x - a)^2 + y^2 = a^2$

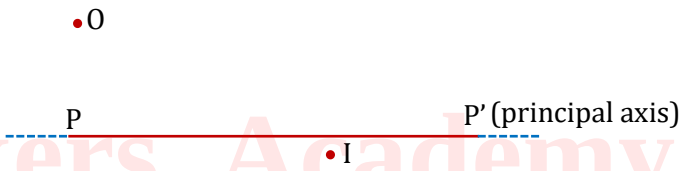
(B) The path of the light ray inside medium is a parabola given by $(x - a)^2 = y$

(C) The deviation of ray from its initial direction is $\delta = \pi/3$

(D) Ray emerges grazing at the surface.

PGO177

8. The location *I* of image for object location *O* can be obtained by placing



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(A) Concave mirror left of *O* facing towards *O*.

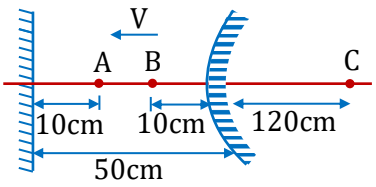
(B) Concave mirror right of *I* facing towards *I*.

(C) Convex mirror between *O* and I facing towards *O*.

(D) Convex mirror right of *I* facing away from *I*.

PGO151

9. In the figure shown consider the first reflection at the plane mirror and second at the convex mirror. *AB* is object.



(A) the second image is real, inverted of 1/5 th magnification w.r.t *AB*

(B) the second image is virtual and erect with magnification 1/5 w.r.t *AB*

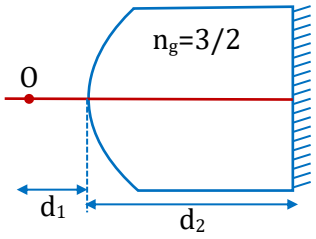
(C) the second image moves towards the convex mirror

(D) the second image moves away from the convex mirror.

PGO156

For More Join: @IITJEE_Advanced

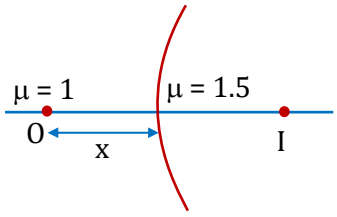
10. In the figure shown a point object *O* is placed in air on the principal axis. The radius of curvature of the spherical surface is 60 *cm*. I_f is the final image formed after all the refractions and reflections.



(A) If d_1 = 120 *cm*, then the 'I_f' is formed on '*O*' for any value of d_2.
(B) If d_1 = 240 *cm*, then the 'I_f' is formed on '*O*' only if d_2 = 360 *cm*.
(C) If d_1 = 240 *cm*, then the 'I_f' is formed on '*O*' for all values of d_2.
(D) If d_1 = 240 *cm*, then the 'I_f' cannot be formed on '*O*'.

PGO194

11. A convex spherical surface of radius 10 *cm* is used to form the image of a point object as shown. What should be the object distance *x* so that the distance between object and the image is 1 *m* ?



(A) 40 *cm* (B) 50 *cm* (C) 60 *cm* (D) 70 *cm*

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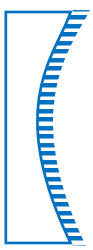
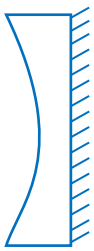
PGO197

12. A lens is formed by a material having refracting index $\frac{3}{2}$ and radii of curvature 20*cm* and 10 *cm*. Then choose the correct option(s) :-

(A) If lens is concavo-convex and light is falling on the surface having radius of curvature 20 *cm*, then focal length is 40 *cm*
(B) If lens is concavo-convex and light is falling on the surface having radius of curvature 10 *cm*, then focal length is –40 *cm*
(C) If lens is convexo-concave and light is falling on the surface having radius of curvature 20 *cm*, then focal length is 40 *cm*
(D) If lens is convexo-concave and light is falling on the surface having radius of curvature 10 *cm*, then focal length is –40 *cm*

PGO190

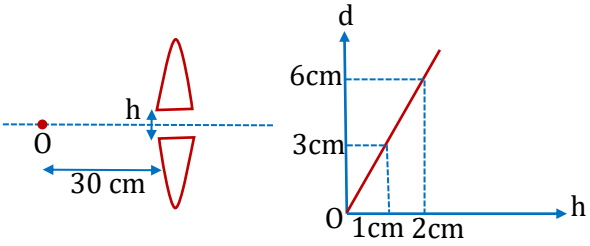
13. Which of the following silvered lenses kept in air may form real image of a real object.

(A) 
(B) 
(C) 
(D) 

PGO199

For More Join: @IITJEE_Advanced

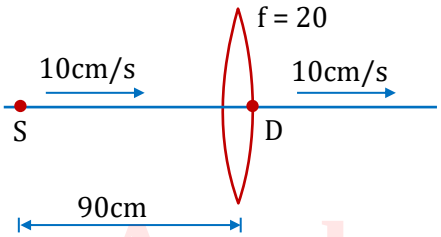
14. Figure shows a convex lens cut symmetrically into two equal halves and separated laterally by a distance *h*. A point object *O* placed symmetrically at a distance 30 *cm*, from the lens halves, forms two images separated by a distance *d*. A plot of *d* versus *h* is shown in figure. The focal length of the lens is :-



(A) 22.5 *cm* (B) 40 *cm* (C) 45 *cm* (D) 20 *cm*

PGO201

15. The figure shows initial positions of a source of light *S* and a light detector *D*. Both *S* and *D* move with velocity 10 *cm/s*, as shown. The converging lens of focal length 20 *cm* is fixed. Time instant(s) when the detector detects maximum intensity of light is/are :-



(A) 3 sec (B) 4.5 sec (C) 6 sec (D) equal intensity at all time

PGO208

16. In displacement method, the distance between object and screen is 96 *cm*. The ratio of length of two images formed by a convex lens placed between them is 4.84 :-

(A) Ratio of the length of object to the length of shorter image is 11/5.

(B) Distance between the two positions of the lens is 36 *cm*.

(C) Focal length of the lens is 20.625 *cm*.

(D) Distance of the lens from the shorter image is 30 *cm*.

PGO217

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 17 TO 19

A turnip sits before a thin converging lens, outside the focal point of the lens. The lens is filled with a transparent gel so that it is flexible; by squeezing its ends toward its center [as indicated in figure(*a*)], you can change the curvature of its front and rear sides.

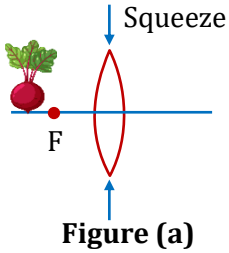


Figure (a)

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17. When you squeeze the lens, the image.
(A) moves towards the lens (B) moves away from the lens
(C) shifts up (D) remains as it is

PGO210

18. When you squeeze the lens, the lateral height of image.
(A) increases (B) decreases (C) remains same (D) data insufficient

PGO211

19. Suppose that a sharp image must be formed on a card which is at a certain distance behind the lens [figure(*b*)], while you move the turnip away from the lens, then you should

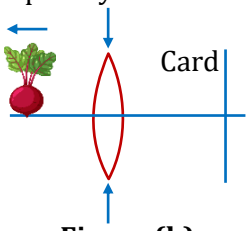


Figure (b)

(A) decrease the squeeze of the lens (B) increase the squeeze of the lens
(C) keep the card and lens as it is. (D) move the card away from the lens

PGO212

PARAGRAPH FOR QUESTION NO. 20 TO 23

20. If an astronomical telescope has objective and eye-pieces of focal lengths 200 *cm* and 4 *cm* respectively, then the magnifying power of the telescope for the normal vision is
(A) 42 (B) 50 (C) 58 (D) 204

PGO238

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21. In the above question the length of the telescope for normal vision, is
(A) 204 *cm* (B) 200 *cm* (C) 196 *cm* (D) 203.45 *cm*

PGO239

22. In the above question magnifying power of the telescope for distinct vision is
(A) 42 (B) 50 (C) 58 (D) 204

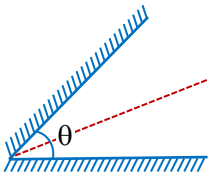
PGO240

23. In the above question the length of the telescope for distinct vision is
(A) 204 *cm* (B) 200 *cm* (C) 196 *cm* (D) 203.45 *cm*

PGO241

MATRIX MATCH TYPE QUESTION

24. Angle between two mirrors 'θ' and location of object is given in column *I* and some possible number of images are given in column II.



	Column-I		**Column-II**
(A)	θ = 30° and object is on bisector	(P)	9
(B)	θ = 36° and object is on bisector	(Q)	11
(C)	θ = 22.5° and object is on bisector	(R)	8
(D)	θ = 50° and object is on bisector	(S)	15

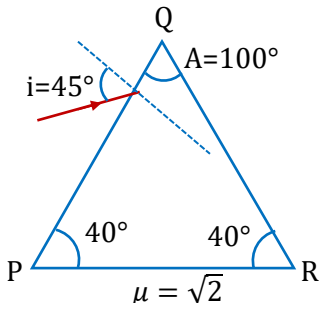
PGO129

For More Join: @IITJEE_Advanced

25. Light is incident at surface *PQ* of prism as shown in column-I then match the column-I with column-II (surrounding medium is air in all cases)

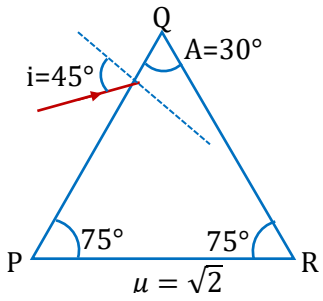
Column-I **Column-II**

(A)



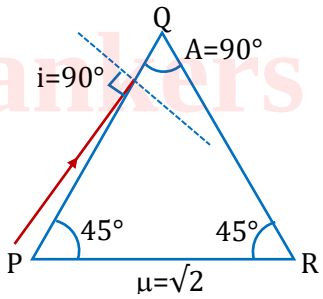
(P) Total internal reflection takes place at surface *QR*

(B)



(Q) Light emerges normally from the surface *QR*

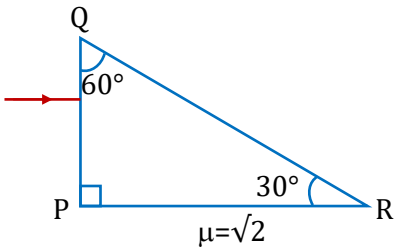
(C)



Grazing incidence

(R) Light emerges parallel to surface *QR*

(D)



Normal incidence

(S) The light ray emerges from face perpendicularly

(T) When light ray passes through the prism it is parallel to the base *PR*.

PGO187

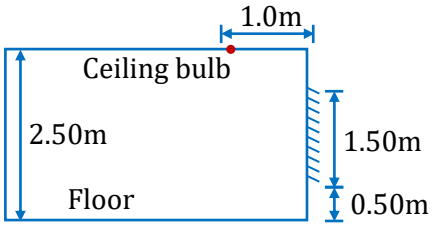
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EXERCISE (S)

Plane Mirror

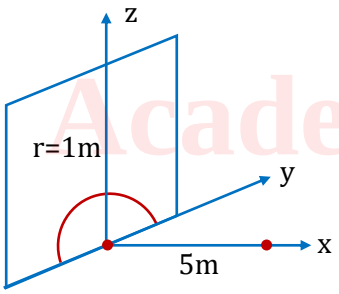
1. A plane mirror of length 1.50 m hangs on a 2.50 m tall wall inside a room. The bottom of mirror is 0.50 m above the floor. A small bulb is situated at the ceiling of room at a distance of 1.0 m from the wall of plane mirror as shown in the figure. Find the length (in m) of patch of light on the floor due to reflection from mirror (assume floor size is sufficient)



PGO246

2. A plane mirror of semicircular shape (radius = 1m) is placed in y-z plane with its diameter along y-axis and centre at origin. A point object in the form of light source is placed on the axis of mirror at point (5, 0, 0). At a distance of 10 m from plane mirror there is a wall also parallel to y-z plane. The ratio of area of spot of light on wall to that of mirror is

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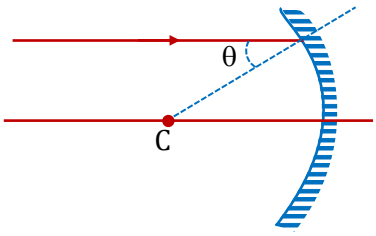
PGO247

3. A point object is moving with velocity $6\hat{\imath} + \hat{\jmath} + 2\hat{k}$ (m/s) in front of a plane mirror whose reflecting side is xz-plane. If the mirror also moves with a velocity $\hat{\imath} - \hat{\jmath}$ (m/s), what is the speed of the image in m/s rounded off to nearest integer ?

PGO248

Spherical Mirror

4. A ray incident parallel to principal axis on a concave mirror at an angle θ after two reflection, reflected ray becomes parallel to incident ray. If $\theta = \frac{\pi}{N}$ (in radian) then find the value of N.



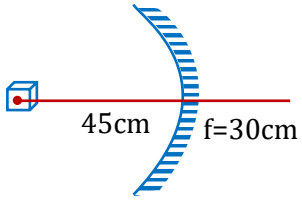
PGO139

For More Join: @IITJEE_Advanced

5. An experimentalist devises a method for finding the radius of curvature of a convex mirror. He uses a plane mirror strip between the object and the convex mirror and adjusts it till the two virtual images formed by reflection at both the mirrors coincide without parallax. In his observations, the object distance from the convex mirror is 0.5 *m* while it is 0.30 *m* in front of the plane mirror. Find the radius of curvature (in *cm*) of the convex mirror.

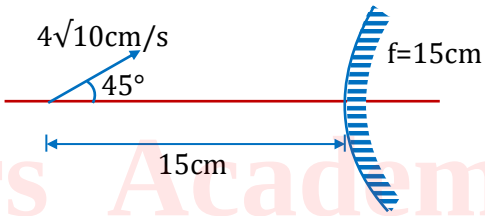
PGO152

6. A cube of side length 1*mm* is placed on the axis of a concave mirror at a distance of 45 *cm* from the pole as shown in the figure. One edge of the cube is parallel to the axis. The focal length of the mirror is 30 *cm*. Find approximate volume of the image.



PGO153

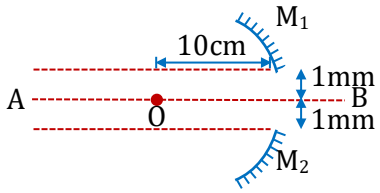
7. In the figure shown, the speed *cm/s* of image with respect to mirror is :



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PGO154

8. A concave mirror of focal length 20 *cm* is cut into two parts from the middle and the two parts are moved perpendicularly by a distance 1 *mm* from the previous principal axis *AB*. If an object is placed of a distance of 10 *cm* from the mirror on the line *AB* then, find the distance between the images formed by the two parts?



PGO157

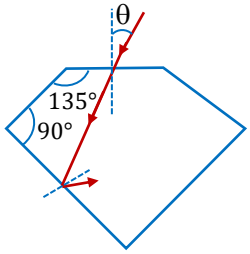
Refraction from Plane Surface

9. A cylindrical bucket of depth 60 *cm* is partly filled with a liquid of refractive index 1.5 and with oil (on top of liquid) of refractive index 2. It appears that the volume of air, volume of liquid and volume of oil are equal, to an observer who views from top of the bucket. The apparent depth of the bucket as seen by the observer is given as α *cm*. Fill $\left(\frac{\alpha}{5}\right)$ in OMR sheet.

PGO163

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10. A ray of light enters a diamond (n = 2) from air and is being internally reflected near the bottom as shown in the figure. Find maximum value of angle θ possible ?



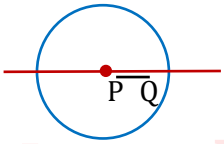
PGO170

11. A room contains air in which the speed of sound is 340 *m/s*. The walls of the room are made of concrete, in which the speed of sound is 1700 *m/s*. (a) Find the critical angle for total internal reflection of sound at the concrete–air boundary. (b) In which medium must the sound be traveling to undergo total internal reflection? (Assume sound waves follow snell's law)

PGO175

Refraction through Curved Surface

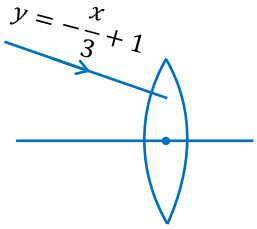
12. A small object of length 1 *mm* lies along the principal axis of a spherical glass of radius R = 10 *cm* and refractive index is 3/2. The object is seen from air along the principal axis from left. The distance of object from the centre is 5 *cm*. Find the size of the image. Is real, inverted ?



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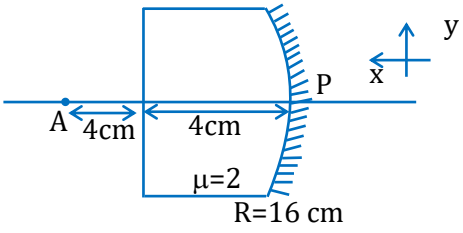
PGO195

13. A convex lens of focal length 15 *cm* is placed at the origin with its optical axis coinciding with the x-axis. A ray of light given by $y = -\frac{x}{3}+1$ is incident on the lens. The equation of refracted ray is $y = -0.1mx + c$. (x and y are in *cm*). Find the value of m.



PGO250

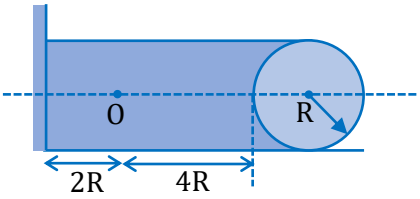
14. A curved thick glass surface is silvered at curved face & not silvered on plane surface. Object is placed at A as shown in figure. Considering P (pole of the silvered surface) as origin. If x – co–ordinate of final image is 'n' *cm* then find n.



PGO251

For More Join: @IITJEE_Advanced

15. The figure shows a very long circular cylinder of radius $R = 2m$ made of glass $\mu_g = \frac{3}{2}$ fixed to the horizontal surface. The space between the cylinder and a vertical wall (plane mirror) is filled with water of refractive index $\mu_w = 4/3$, as shown in figure. Find the positions of the final image formed of an object O placed on the horizontal axis passing through the centre of mass C of the cylinder as shown in figure. Express your answer with respect to centre C of the cylinder. Consider only paraxial rays lying in the vertical plane containing the point O and C.



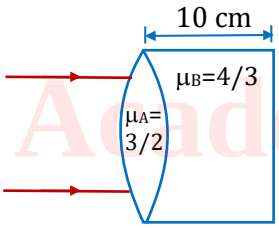
PGO252

Lens

16. A plano convex lens ($\mu = 1.5$) has a maximum thickness of 1 *mm*. If diameter of its aperture is 4 *cm*. Find (i) Radius of curvature of curved surface; (ii) its focal length in air.

PGO188

17. A parallel paraxial beam of light is incident on the arrangement as shown $\mu_A = 3/2$, $\mu_B = 4/3$, the two spherical surfaces are very close and each has radius of curvature 10 *cm*. Find the point where the rays are focussed. (w.r.t. point of entry)



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PGO198

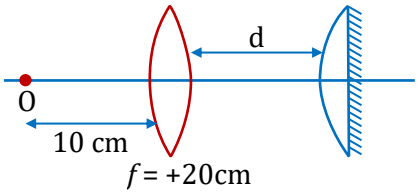
18. A converging beam of rays is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 *cm* from the lens. If the lens is removed, the point where the rays meet will move 5 *cm* closer to the mounting that holds. Find the focal length (in *cm*) of the lens without sign.

PGO202

19. A point source of light is kept at a distance of 150 *cm* from a converging lens, on its optical axis. The focal length of the lens is 100 *cm* and its diameter is 3 *cm*. A screen is placed on the other side of the lens, perpendicular to the axis of lens, at a distance 200 *cm* from it. Then find the area of the illuminated part of the screen? (Assume all rays incident on lens as paraxial)

PGO203

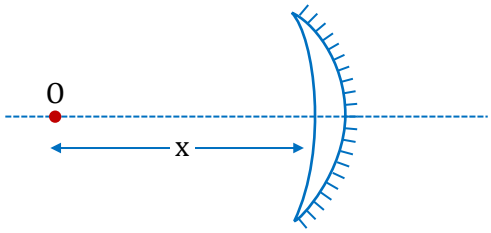
20. A convex lens of focal length 20 *cm* and another plano convex lens of focal length 40 *cm* are placed co-axially (see fig.). The plano convex lens is silvered on plane surface. What should be the $5d$ (in *m*) so that final image of the object '*O*' is formed on *O* itself.



PGO204

For More Join: @IITJEE_Advanced

21. Radii of curvature of a concavo-convex lens (refractive index = 1.5) are 40 *cm* and 20 *cm* as shown. The convex side is silvered. The distance *x* (in *cm*) on the principal axis where an object is placed so that its image is created on the object itself, is given as 4β. Find the value of β.



PGO205

Prism

22. A flint glass convex lens of focal length 16 *cm* is placed in contact with crown glass lens. The refractive indices of crown glass for violet and red colour are $\mu_v = 1.525$, $\mu_r = 1.515$ and for flint glass $\mu_{v'} = 1.655$ and $\mu_{r'} = 1.645$.

(a) Find the nature and focal length of the crown glass lens which will form an achromatic combination with the flint glass lens.

(b) What is the focal length of the combination?

PGO223

Defects of Eye and Optical Instruments

23. A simple telescope consisting of an objective of focal length 60 *cm* and a single eye lens of focal length 5 *cm* is focussed on a distant object in such a way that parallel rays emerge from the eye lens. If the object subtend an angle of 2° at the objective then find the angular width of the image (in degree).

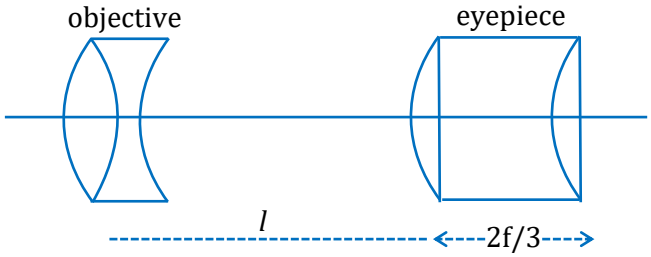
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PGO253

24. You are trying to construct a compound microscope given two lenses with focal lengths $f_1 = 1.0$ *cm* and $f_2 = 4.0$ *cm.* How far apart (in *cm*) should you place the lenses in order to obtain an angular magnification of 60 ? (Assume normal adjustment of microscope)

PGO254

25. The objective of an astronomical telescope consists of two thin lenses in contact, of focal lengths +20 *cm* and -25 *cm* respectively. Eyepiece of the same telescope consists of two plano convex lenses each of focal length *f* separated by 2/3 *f* as shown in the figure. Find the value of *l* for which final image will be formed at infinity with its angular magnification 100/3. Also find *f*.



PGO255

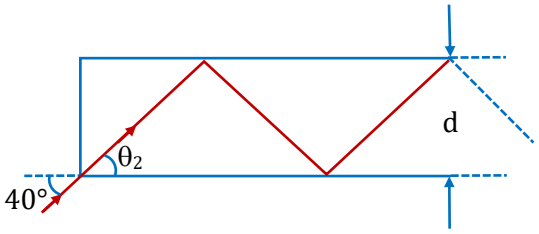
26. An object viewed from a near point distance of 25 *cm*, using a microscopic lens with magnification '6', gives an unresolved image. A resolved image is observed at infinite distance with a total magnification double the earlier using an eyepiece along with the given lens and a tube of length 0.6 *m*. The focal length of the eyepiece is equal to _____*cm*.

PGO256

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. In figure, the optical fiber is $\ell = 2m$ long and has a diameter of $d = 20\ \mu m$. If a ray of light is incident on one end of the fiber at angle $\theta_1 = 40°$, the number of reflection it makes before emerging from the other end is close to: (Refractive index of fibre is 1.31 and sin 40° = 0.64) **[JEE (Main) 2019]**

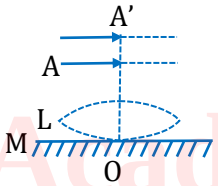


(1) 55000 (2) 57000 (3) 66000 (4) 45000

PGO042

2. A thin convex lens *L* (refractive index = 1.5) is placed on a plane mirror *M*. When a pin is placed at *A*, such that *OA* = 18 *cm*, its real inverted image is formed at *A* itself, as shown in figure. When a liquid of refractive index μ_1 is put between the lens and the mirror. The pin has to be moved to *A*', such that *OA*' = 27 *cm*, to get its inverted real image at *A*' itself. The value of μ_1 will be :-

[JEE (Main) 2019]

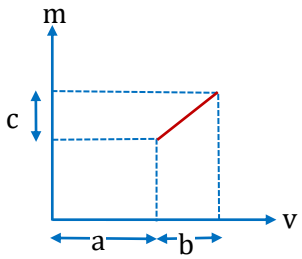


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(1) $\sqrt{2}$ (2) $\frac{4}{3}$ (3) $\sqrt{3}$ (4) $\frac{3}{2}$

PGO043

3. The graph shows how the magnification *m* produced by a thin lens varies with image distance *v*. What is the focal length of the lens used ? **[JEE (Main) 2019]**



(1) $\frac{b^2c}{a}$ (2) $\frac{b^2}{ac}$ (3) $\frac{a}{c}$ (4) $\frac{b}{c}$

PGO044

4. An object is at a distance of 20 *m* from a convex lens of focal length 0.3 *m*. The lens forms an image of the object. If the object moves away from the lens at a speed of 5 *m/s*, the speed and direction of the image will be : **[JEE (Main) 2019]**

(1) 0.92×10^{-3} *m/s* away from the lens (2) 2.26×10^{-3} *m/s* away from the lens

(3) 1.16×10^{-3} *m/s* towards the lens (4) 3.22×10^{-3} *m/s* towards the lens

PGO046

For More Join: @IITJEE_Advanced

5. In a compound microscope, the magnified virtual image is formed at a distance of 25 *cm* from the eye-piece. The focal length of its objective lens is 1 *cm*. If the magnification is 100 and the tube length of the microscope is 20 *cm*, then the focal length of the eye-piece lens (in *cm*) is _____.

[JEE (Main) 2020]

PGO047

6. The distance between an object and a screen is 100 *cm*. A lens can produce real image of the object on the screen for two different positions between the screen and the object. The distance between these two positions is 40 *cm*. If the power of the lens is close to $\left(\frac{N}{100}\right)D$ where N is an integer, the value of N is _____.

[JEE (Main) 2020]

PGO048

7. If we need a magnification of 375 from a compound microscope of tube length 150 *mm* and an objective of focal length 5 *mm*, the focal length of the eye-piece, should be close to :

[JEE (Main) 2020]

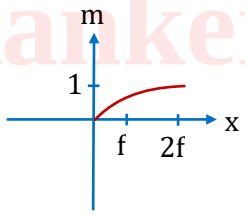
(1) 22 *mm* (2) 12 *mm* (3) 33 *mm* (4) 2 *mm*

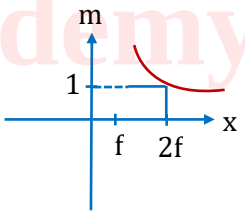
PGO049

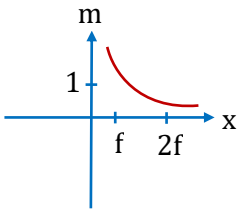
8. An object is gradually moving away from the focal point of a concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (m) versus distance of the object from the mirror (x) is correctly given by :

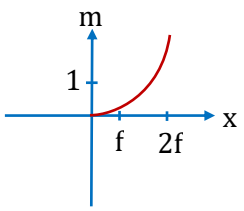
(Graphs are drawn schematically and are not to scale) **[JEE (Main) 2020]**

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(1)


(2)


(3)


(4)


PGO050

9. There is a small source of light at some depth below the surface of water (refractive index = $\frac{4}{3}$) in a tank of large cross sectional surface area. Neglecting any reflection from the bottom and absorption by water, percentage of light that emerges out of surface is (nearly) :

[Use the fact that surface area of a spherical cap of height h and radius of curvature r is $2\pi rh$]:

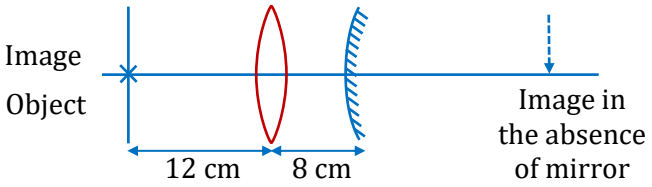
[JEE (Main) 2020]

(1) 17% (2) 21% (3) 34% (4) 50%

PGO051

For More Join: @IITJEE_Advanced

10. An object is placed at a distance of 12 *cm* from a convex lens. A convex mirror of focal length 15 *cm* is placed on other side of lens at 8 *cm* as shown in the figure. Image of object coincides with the object.



When the convex mirror is removed, a real and inverted image is formed at a position. The distance of the image from the object will be .....(*cm*) **[JEE (Main) 2021]**

PGO054

11. Car *B* overtakes another car *A* at a relative speed of 40 ms^{-1}. How fast will the image of car *B* appear to move in the mirror of focal length 10 *cm* fitted in car *A*, when the car *B* is 1.9 *m* away from the car *A*? **[JEE (Main) 2021]**

(1) 4 ms^{-1} (2) 0.2 ms^{-1} (3) 40 ms^{-1} (4) 0.1 ms^{-1}

PGO055

12. A ray of light is incident at an angle of incidence 60° on the glass slab of refractive index $\sqrt{3}$. After refraction, the light ray emerges out from other parallel faces and lateral shift between incident ray and emergent ray is $4\sqrt{3}$ *cm*. The thickness of the glass slab is _____ *cm*. **[JEE (Main) 2022]**

PGO056

13. Two identical thin biconvex lenses of focal length 15 *cm* and refractive index 1.5 are in contact with each other. The space between the lenses is filled with a liquid of refractive index 1.25. The focal length of the combination is _____ *cm*. **[JEE (Main) 2022]**

PGO057

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14. A thin cylindrical rod of length 10 *cm* is placed horizontally on the principal axis of a concave mirror of focal length 20 *cm*. The rod is placed in a such a way that mid-point of the rod is at 40 *cm* from the pole of mirror. The length of the image formed by the mirror will be $\frac{x}{3}$ cm. The value of *x* is _____.

[JEE (Main) 2023]

PGO259

15. A monochromatic light wave with wavelength λ_1 and frequency v_1 in air enters another medium. If the angle of incidence and angle of refraction at the interface are 45° and 30° respectively, then the wavelength λ_2 and frequency v_2 of the refracted wave are : **[JEE (Main) 2023]**

(1) $\lambda_2 = \lambda_1, \nu_2 = \sqrt{2}\nu_1$ (2) $\lambda_2 = \frac{1}{\sqrt{2}}\lambda_1, \nu_2 = \nu_1$

(3) $\lambda_2 = \sqrt{2}\lambda_1, \nu_2 = \nu_1$ (4) $\lambda_2 = \lambda_1, \nu_2 = \frac{1}{\sqrt{2}}\nu_1$

PGO260

16. In a reflecting telescope, a secondary mirror is used to: **[JEE (Main) 2023]**

(1) reduce the problem of mechanical support
(2) remove spherical aberration
(3) make chromatic aberration zero
(4) move the eyepiece outside the telescopic tube

PGO261

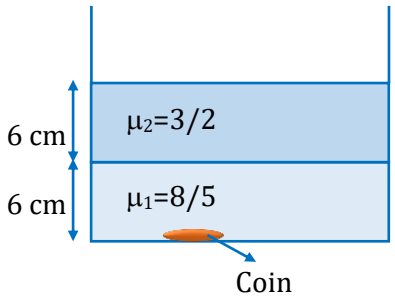
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17. A convex mirror of radius of curvature 30 *cm* forms an image that is half the size of the object. The object distance is : **[JEE (Main) 2024]**

(1) –15 *cm* (2) 45 *cm* (3) –45 *cm* (4) 15 *cm*

PGO271

18. Two immiscible liquids of refractive indices $\frac{8}{5}$ and $\frac{3}{2}$ respectively are put in a beaker as shown in the figure. The height of each column is 6 *cm*. A coin is placed at the bottom of the beaker. For near normal vision, the apparent depth of the coin is $\frac{\alpha}{4}$ *cm*. The value of α is_____. **[JEE (Main) 2024]**

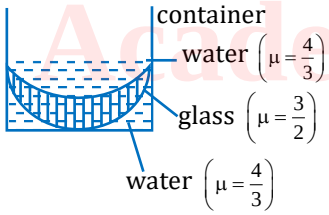


PGO272

19. In the diagram given below, there are three lenses formed. Considering negligible thickness of each of them as compared to $[R_1]$ and $[R_2]$, i.e., the radii of curvature for upper and lower surfaces of the glass lens, the power of the combination is

[JEE (Main) 2025]

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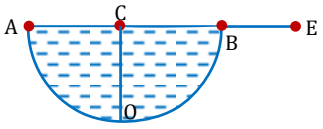
(1) $-\frac{1}{6}\left(\frac{1}{|R_1|}+\frac{1}{|R_2|}\right)$ (2) $-\frac{1}{6}\left(\frac{1}{|R_1|}-\frac{1}{|R_2|}\right)$

(3) $\frac{1}{6}\left(\frac{1}{|R_1|}+\frac{1}{|R_2|}\right)$ (4) $\frac{1}{6}\left(\frac{1}{|R_1|}-\frac{1}{|R_2|}\right)$

PGO336

20. A hemispherical vessel is completely filled with a liquid of refractive index μ. A small coin is kept at the lowest point (O) of the vessel as shown in figure. The minimum value of the refractive index of the liquid so that a person can see the coin from point E (at the level of the vessel) is_____.

[JEE (Main) 2025]



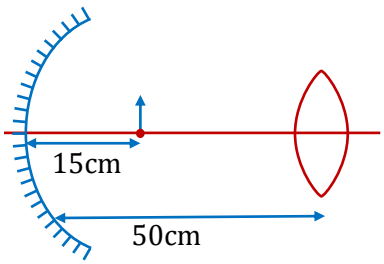
(1) $\sqrt{3}$ (2) $\frac{3}{2}$ (3) $\sqrt{2}$ (4) $\frac{\sqrt{3}}{2}$

PGO337

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

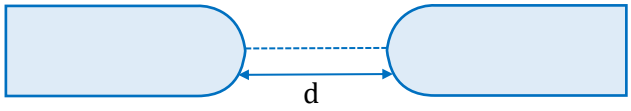
1. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 *cm* each, separated by a distance of 50 *cm* in air (refractive index = 1) as shown in the figure. An object is placed at a distance of 15 *cm* from the mirror. Its erect image formed by this combination has magnification M_1. When the set-up is kept in a medium of refractive index 7/6 the magnification becomes M_2. The magnitude $\left|\frac{M_2}{M_1}\right|$ is



[JEE (Advanced) 2015]

PGO079

2. Two identical glass rods S_1 and S_2 (refractive index = 1.5) have one convex end of radius of curvature 10 cm. They are placed with the curved surfaces at a distance *d* as shown in the figure, with their axes (shown by the dashed line) aligned. When a point source of light *P* is placed inside rod S_1 on its axis at a distance of 50 cm from the curved face, the light rays emanating from it are found to be parallel to the axis inside S_2. The distance *d* is : **[JEE (Advanced) 2015]**

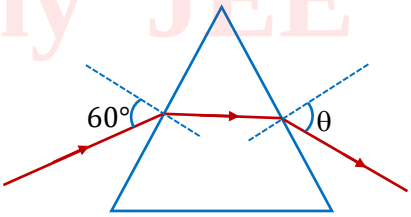


(A) 60 *cm* (B) 70 *cm* (C) 80 *cm* (D) 90 *cm*

PGO080

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3. A monochromatic beam of light is incident at 60° on one face of an equilateral prism of refractive index *n* and emerges from the opposite face making an angle θ(*n*) with the normal (see the figure). For $n = \sqrt{3}$ the value of θ is 60° and $\frac{d\theta}{dn} = m$. The value of *m* is

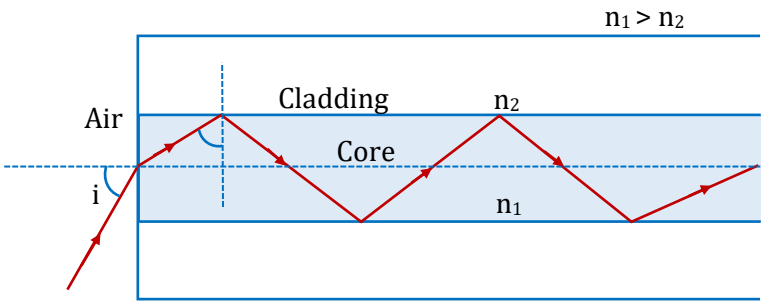


[JEE (Advanced) 2015]

PGO081

PARAGRAPH FOR QUESTION NO. 4 AND 5

Light guidance in an optical fiber can be understood by considering a structure comprising of thin solid glass cylinder of refractive index n_1 surrounded by a medium of lower refractive index n_2. The light guidance in the structure takes place due to successive total internal reflections at the interface of the media n_1 and n_2 as shown in the figure. All rays with the angle of incidence *i* less than a particular value of i_m are confined in the medium of refractive index n_1. The numerical aperture (NA) of the structure is defined as sin i_m. **[JEE (Advanced) 2015]**



For More Join: @IITJEE_Advanced

4. For two structures namely S_1 with $n_1 = \sqrt{45}/4$ and $n_2 = 3/2$, and S_2 with $n_1 = 8/5$ and $n_2 = 7/5$ and taking the refractive index of water to be $4/3$ and that of air to be 1, the correct option(s) is (are)

(A) NA of S_1 immersed in water is the same as that of S_2 immersed in liquid of refractive index $\frac{16}{3\sqrt{15}}$.

(B) NA of S_1 immersed in liquid of refractive index $\frac{6}{\sqrt{15}}$ is the same as that of S_2 immersed in water.

(C) NA of S_1 placed in air is the same as that of S_2 immersed in liquid of refractive index $\frac{4}{\sqrt{15}}$.

(D) NA of S_1 placed in air is the same as that of S_2 placed in water.

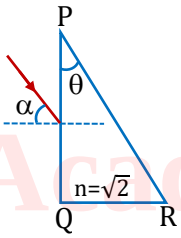
PGO082

5. If two structures of same cross-sectional area, but different numerical apertures NA_1 and NA_2 ($NA_2 < NA_1$) are joined longitudinally, the numerical aperture of the combined structure is

(A) $\frac{NA_1 NA_2}{NA_1 + NA_2}$ (B) $NA_1 + NA_2$ (C) NA_1 (D) NA_2

PGO083

6. A parallel beam of light is incident from air at an angle α on the side PQ of a right angled triangular prism of refractive index $n = \sqrt{2}$. Light undergoes total internal reflection in the prism at the face PR when α has a minimum value of 45°. The angle θ of the prism is : **[JEE (Advanced) 2016]**

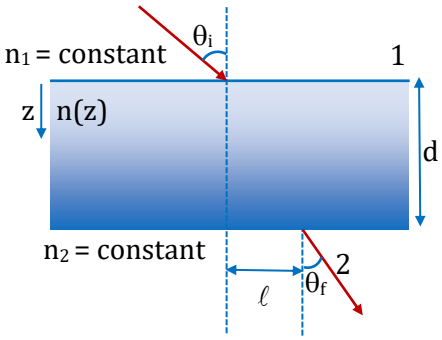


Rankers Academy JEE

(A) 15° (B) 22.5° (C) 30° (D) 45°

PGO084

7. A transparent slab of thickness d has a refractive index $n(z)$ that increases with z. Here z is the vertical distance inside the slab, measured from the top. The slab is placed between two media with uniform refractive indices n_1 and n_2 ($> n_1$), as shown in the figure. A ray of light is incident with angle θ_i from medium 1 and emerges in medium 2 with refraction angle θ_f with a lateral displacement ℓ. Which of the following statement(s) is(are) true ? **[JEE (Advanced) 2016]**



(A) ℓ is independent of n_2 (B) ℓ is dependent on $n(z)$

(C) $n_1 \sin\theta_i = (n_2 - n_1)\sin\theta_f$ (D) $n_1 \sin\theta_i = n_2 \sin\theta_f$

PGO085

For More Join: @IITJEE_Advanced

8. A plano-convex lens is made of a material of refractive index n. When a small object is placed 30 cm away in front of the curved surface of the lens, an image of double the size of the object is produced. Due to reflection from the convex surface of the lens, another faint image is observed at a distance of 10 cm away from the lens. Which of the following statement(s) is(are) true?

[JEE (Advanced) 2016]

(A) The refractive index of the lens is 2.5

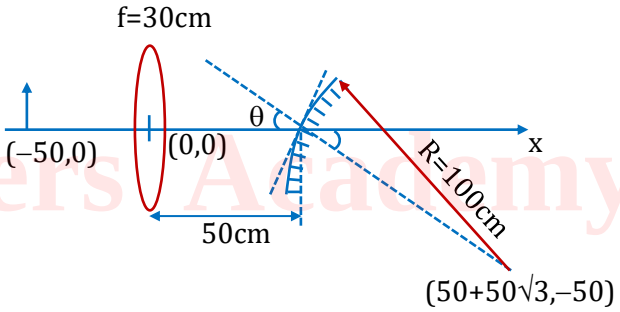
(B) The radius of curvature of the convex surface is 45 cm

(C) The faint image is erect and real

(D) The focal length of the lens is 20 cm.

PGO086

9. A small object is placed 50 cm to the left of thin convex lens of focal length 30 cm. A convex spherical mirror of radius of curvature 100 cm is placed to the right of the lens at a distance of 50 cm. The mirror is tilted such that the axis of the mirror is at an angle $\theta = 30°$ to the axis of the lens, as shown in the figure. If the origin of the coordinate system is taken to be at the centre of the lens, the coordinates (in cm) of the point (x, y) at which the image is formed are : **[JEE (Advanced) 2016]**



(A) $(25, 25\sqrt{3})$ (B) $\left(\frac{125}{3}, \frac{25}{\sqrt{3}}\right)$ (C) $(50-25\sqrt{3}, 25)$ (D) (0, 0)

PGO087

10. For an isosceles prism of angle A and refractive index μ, it is found that the angle of minimum deviation $\delta_m = A$. Which of the following options is/are correct ? **[JEE (Advanced) 2017]**

(A) At minimum deviation, the incident angle i_1 and the refracting angle r_1 at the first refracting surface are related by $r_1 = (i_1/2)$

(B) For this prism, the refractive index μ and the angle of prism A are related as $A = \frac{1}{2}\cos^{-1}\left(\frac{\mu}{2}\right)$

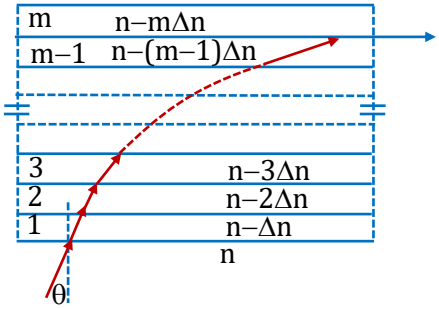
(C) For this prism, the emergent ray at the second surface will be tangential to the surface when the angle of incidence at the first surface is $i_1 = \sin^{-1}\left[\sin A\sqrt{4\cos^2\frac{A}{2}-1}-\cos A\right]$

(D) For the angle of incidence $i_1 = A$, the ray inside the prism is parallel to the base of the prism.

PGO088

For More Join: @IITJEE_Advanced

11. A monochromatic light is travelling in a medium of refractive index $n = 1.6$. It enters a stack of glass layers from the bottom side at an angle $\theta = 30°$. The interfaces of the glass layers are parallel to each other. The refractive indices of different glass layers are monotonically decreasing as $n_m = n - m\Delta n$, where n_m is the refractive index of the mth slab and $\Delta n = 0.1$ (see the figure). The ray is refracted out parallel to the interface between the $(m-1)^{th}$ and m^{th} slabs from the right side of the stack. What is the value of m ? **[JEE (Advanced) 2017]**



PGO089

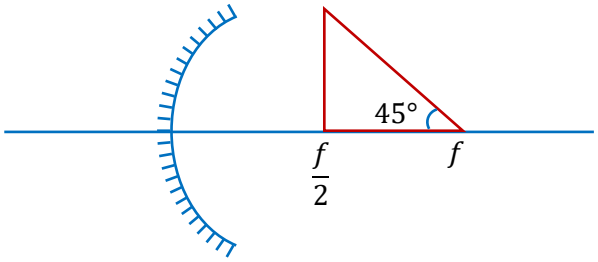
12. Sunlight of intensity 1.3 $kW\ m^{-2}$ is incident normally on a thin convex lens of focal length 20 cm. Ignore the energy loss of light due to the lens and assume that the lens aperture size is much smaller than its focal length. The average intensity of light, in $kW\ m^{-2}$, at a distance 22 cm from the lens on the other side is _____. **[JEE (Advanced) 2018]**

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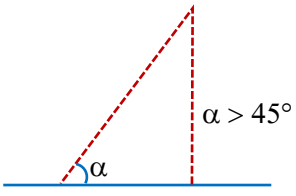
PGO090

13. A wire is bent in the shape of a right angled triangle and is placed in front of a concave mirror of focal length f, as shown in the figure. Which of the figures shown in the four options qualitatively represent(s) the shape of the image of the bent wire ? (These figures are not to scale.)

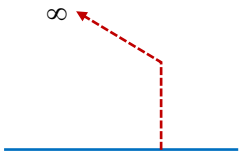
[JEE (Advanced) 2018]



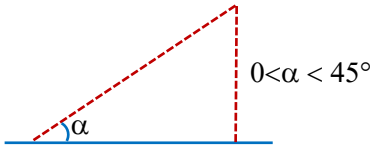
(A)



(B)



(C)



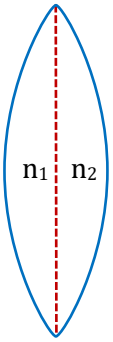
(D)



PGO091

For More Join: @IITJEE_Advanced

14. A thin convex lens is made of two materials with refractive indices n_1 and n_2, as shown in figure. The radius of curvature of the left and right spherical surfaces are equal. f is the focal length of the lens when $n_1 = n_2 = n$. The focal length is $f + \Delta f$ when $n_1 = n$ and $n_2 = n + \Delta n$. Assuming $\Delta n << (n-1)$ and $1 < n < 2$, the correct statement(s) is/are : **[JEE (Advanced) 2019]**



(A) The relation between $\frac{\Delta f}{f}$ and $\frac{\Delta n}{n}$ remains unchanged if both the convex surfaces are replaced by concave surfaces of the same radius of curvature.

(B) $\left|\frac{\Delta f}{f}\right| < \left|\frac{\Delta n}{n}\right|$

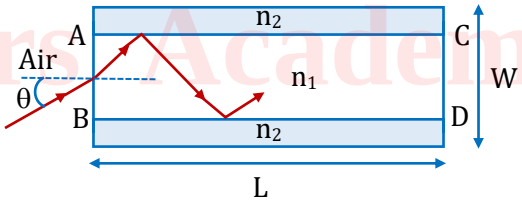
(C) For $n = 1.5$, $\Delta n = 10^{-3}$ and $f = 20$ *cm*, the value of $|\Delta f|$ will be 0.02 *cm* (round off to 2nd decimal place)

(D) If $\frac{\Delta n}{n} < 0$ then $\frac{\Delta f}{f} > 0$

PGO092

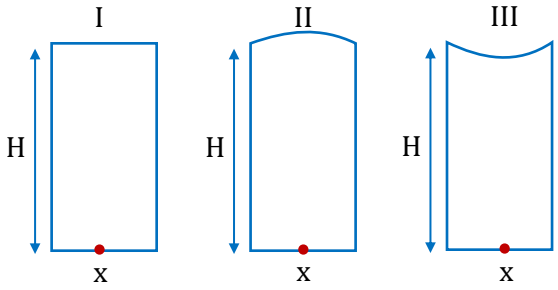
15. A planar structure of length L and width W is made of two different optical media of refractive indices $n_1 = 1.5$ and $n_2 = 1.44$ as shown in figure. If $L >> W$, a ray entering from end AB will emerge from end CD only if the total internal reflection condition is met inside the structure. For $L = 9.6$ *m*, if the incident angle θ is varied, the maximum time taken by a ray to exit the plane CD is $t \times 10^{-9}$ *s*, where t is ___. [Speed of light $c = 3 \times 10^8$ *m/s*] **[JEE (Advanced) 2019]**

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PGO093

16. Three glass cylinders of equal height $H = 30$ *cm* and same refractive index $n = 1.5$ are placed on a horizontal surfaces shown in figure. Cylinder I has a flat top, cylinder II has a convex top and cylinder III has a concave top. The radii of curvature of the two curved tops are same ($R = 3m$). If H_1, H_2 and H_3 are the apparent depths of a point X on the bottom of the three cylinders, respectively, the correct statement(s) is/are **[JEE (Advanced) 2019]**



(A) $H_3 > H_1$

(B) 0.8 cm < $(H_2 - H_1)$ < 0.9 cm

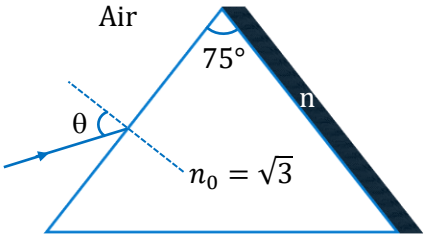
(C) $H_2 > H_3$

(D) $H_2 > H_1$

PGO094

For More Join: @IITJEE_Advanced

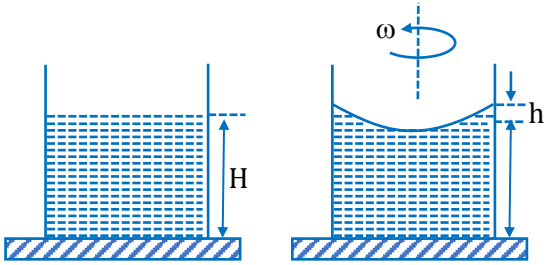
17. A monochromatic light is incident from air on a refracting surface of a prism of angle 75° and refractive index $n_0 = \sqrt{3}$. The other refracting surface of the prism is coated by a thin film of material of refractive index n as shown in figure. The light suffers total internal reflection at the coated prism surface for an incidence angle of $\theta \leq 60°$. The value of n^2 is_____.



[JEE (Advanced) 2019]

PGO095

18. A beaker of radius r is filled with water (refractive index $\frac{4}{3}$) up to a height H as shown in the figure on the left. The beaker is kept on a horizontal table rotating with angular speed ω. This makes the water surface curved so that the difference in the height of water level at the center and at the circumference of the beaker is h ($h << H$, $h << r$), as shown in the figure on the right. Take this surface to be approximately spherical with a radius of curvature R. Which of the following is/are correct? (g is the acceleration due to gravity) **[JEE (Advanced) 2020]**



(A) $R = \frac{h^2 + r^2}{2h}$

(B) $R = \frac{3r^2}{2h}$

(C) Apparent depth of the bottom of the beaker is close to $\frac{3H}{2}\left(1 + \frac{\omega^2 H}{2g}\right)^{-1}$

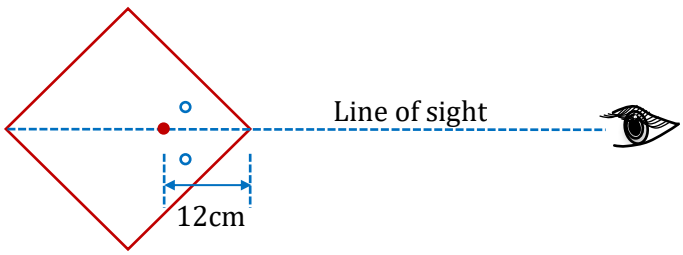
(D) Apparent depth of the bottom of the beaker is close to $\frac{3H}{4}\left(1 + \frac{\omega^2 H}{4g}\right)^{-1}$

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PGO096

19. A large square container with thin transparent vertical walls and filled with water (refractive index $\frac{4}{3}$) is kept on a horizontal table. A student holds a thin straight wire vertically inside the water 12 *cm* from one of its corners, as shown schematically in the figure. Looking at the wire from this corner, another student sees two images of the wire, located symmetrically on each side of the line of sight as shown. The separation (in *cm*) between these images is _____.

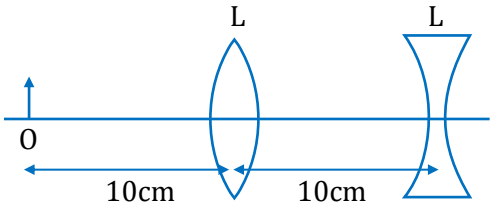
[JEE (Advanced) 2020]



PGO097

For More Join: @IITJEE_Advanced

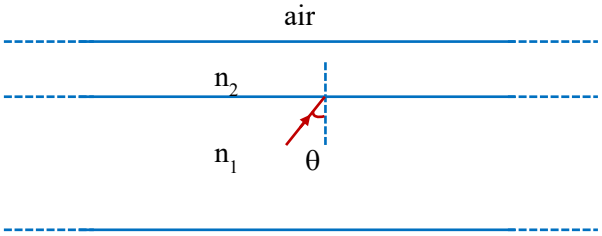
20. An extended object is placed at point O, 10 *cm* in front of a convex lens L_1 and a concave lens L_2 is placed 10 *cm* behind it, as shown in the figure. The radii of curvature of all the curved surfaces in both the lenses are 20 *cm*. The refractive index of both the lenses is 1.5. The total magnification of this lens system is **[JEE (Advanced) 2021]**



(A) 0.4 (B) 0.8 (C) 1.3 (D) 1.6

PGO098

21. A wide slab consisting of two media of refractive indices n_1 and n_2 is placed in air as shown in the figure. A ray of light is incident from medium n_1 to n_2 at an angle θ, where sin θ is slightly larger than $1/n_1$. Take refractive index of air as 1. Which of the following statement(s) is(are) correct? **[JEE (Advanced) 2021]**



(A) The light ray enters air if $n_2 = n_1$

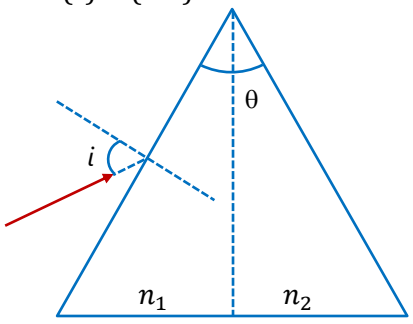
(B) The light ray is finally reflected back into the medium of refractive index n_1 if $n_2 < n_1$

(C) The light ray is finally reflected back into the medium of refractive index n_1 if $n_2 > n_1$

(D) The light ray is reflected back into the medium of refractive index n_1 if $n_2 = 1$

PGO099

22. For a prism of prism angle θ = 60°, the refractive indices of the left half and the right half are, respectively, n_1 and n_2 ($n_2 \geq n_1$) as shown in the figure. The angle of incidence *i* is chosen such that the incident light rays will have minimum deviation if $n_1 = n_2 = n = 1.5$. For the case of unequal refractive indices, $n_1 = n$ and $n_2 = n + \Delta n$ (where $\Delta n << n$), the angle of emergence $e = i + \Delta e$. Which of the following statement(s) is (are) correct ? **[JEE (Advanced) 2021]**



(A) The value of Δe (in radians) is greater than that of Δn

(B) Δe is proportional to Δn

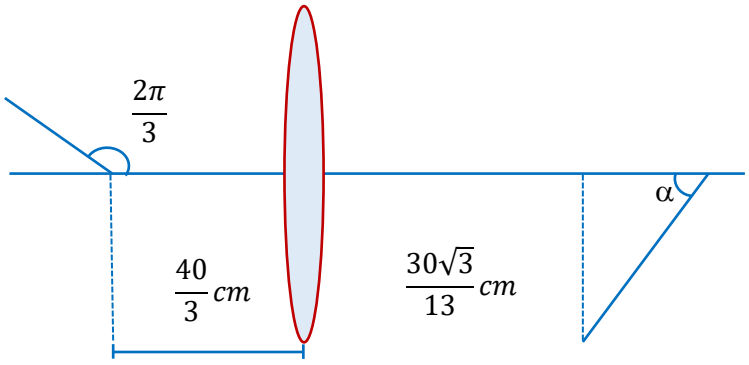
(C) Δe lies between 2.0 and 3.0 milliradians, if $\Delta n = 2.8 \times 10^{-3}$

(D) Δe lies between 1.0 and 1.6 milliradians, if $\Delta n = 2.8 \times 10^{-3}$

PGO100

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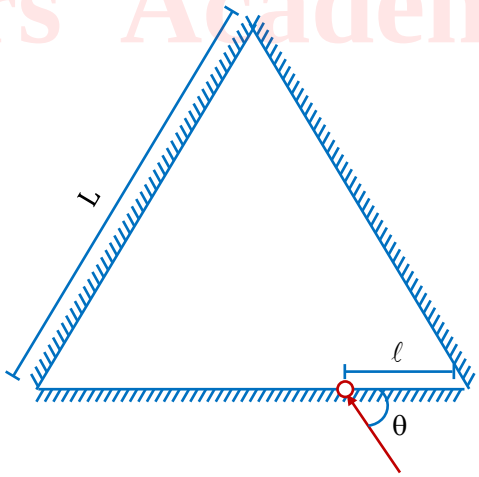
23. A rod of length 2 *cm* makes an angle $\frac{2\pi}{3}$ rad with the principal axis of a thin convex lens. The lens has a focal length of 10 *cm* and is placed at a distance of $\frac{40}{3}$ *cm* from the object as shown in the figure. The height of the image is $\frac{30\sqrt{3}}{13}$ *cm* and the angle made by it with respect to the principal axis is α rad. The value of α is $\frac{\pi}{n}$ *rad*, where *n* is _____. **[JEE (Advanced) 2022]**



PGO101

24. Three plane mirrors form an equilateral triangle with each side of length *L*. There is a small hole at a distance $\ell > 0$ from one of the corners as shown in the figure. A ray of light is passed through the hole at an angle θ and can only come out through the same hole. The cross section of the mirror configuration and the ray of light lie on the same plane. **[JEE (Advanced) 2022]**

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Which of the following statement(*s*) is(are) correct?

(A) The ray of light will come out for θ = 30°, for $0 < \ell < L$.

(B) There is an angle for $\ell = \frac{L}{2}$ at which the ray of light will come out after two reflections.

(C) The ray of light will **NEVER** come out for θ = 60°, and $\ell = \frac{L}{3}$.

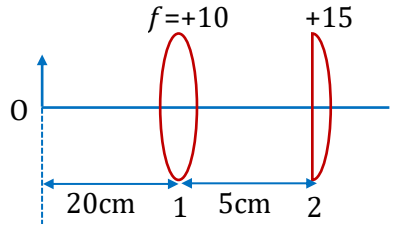
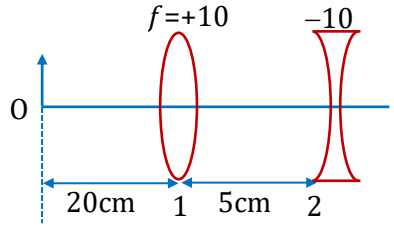
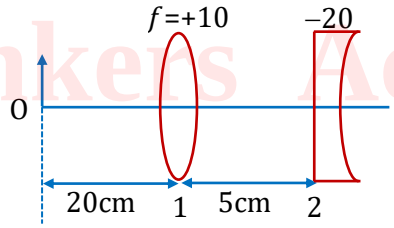
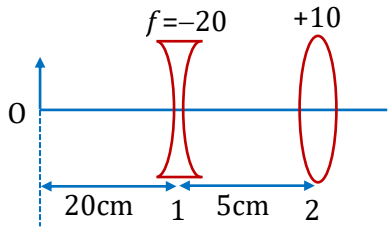
(D) The ray of light will come out for θ = 60°, and $0 < \ell < \frac{L}{2}$ after six reflections.

PGO102

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25. List *I* contains four combinations of two lenses (1 and 2) whose focal lengths (in *cm*) are indicated in the figures. In all cases, the object is placed 20 *cm* from the first lens on the left, and the distance between the two lenses is 5 *cm*. List II contains the positions of the final images.

[JEE (Advanced) 2022]

List-I		List-II	
(I)		**(P)**	Final image is farmed at 7.5 *cm* on the right side of lens 2.
(II)		**(Q)**	Final image is formed at 60.0 *cm* on the right side of lens 2.
(III)		**(R)**	Final image is formed at 30.0 *cm* on the left side of lens 2.
(IV)		**(S)**	Final image is formed at 6.0 *cm* on the right side of lens 2.
		(T)	Final image is formed at 30.0 *cm* on the right side of lens 2.

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Which one of the following options is correct ?

(A) I → P, II → R, III → Q, IV → T

(B) I → Q, II → P, III → T, IV → S

(C) I → P, II → T, III → R, IV → Q

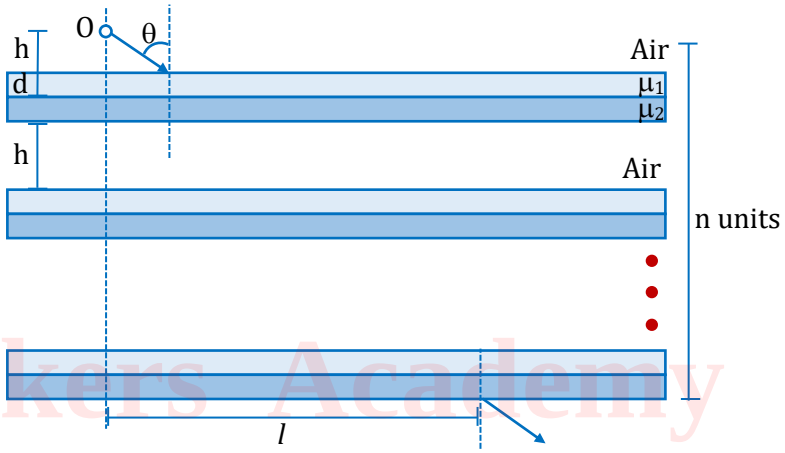
(D) I → T, II → S, III → Q, IV → R

PGO103

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26. Consider a configuration of n identical units, each consisting of three layers. The first layer is a column of air of height $h = \frac{1}{3}\,cm$, and the second and third layers are of equal thickness $d = \frac{\sqrt{3}-1}{2}\,cm$, and refractive indices $\mu_1 = \sqrt{\frac{3}{2}}$ and $\mu_2 = \sqrt{3}$, respectively. A light source O is placed on the top of the first unit, as shown in the figure. A ray of light from O is incident on the second layer of the first unit at an angle of $\theta = 60°$ to the normal. For a specific value of n, the ray of light emerges from the bottom of the configuration at a distance $l = \frac{8}{\sqrt{3}}\,cm$, as shown in the figure. The value of n is _____.

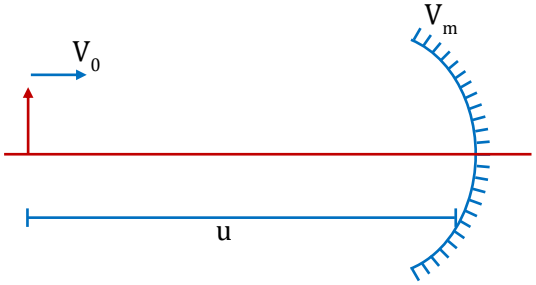
[JEE (Advanced) 2022]



PGO104

27. An object and a concave mirror of focal length $f = 10\ cm$ both move along the principal axis of the mirror with constant speeds. The object moves with speed $V_0 = 15\ cm\ s^{-1}$ towards the mirror with respect to a laboratory frame. The distance between the object and the mirror at a given moment is denoted by u. When $u = 30\ cm$, the speed of the mirror V_m is such that the image is instantaneously at rest with respect to the laboratory frame, and the object forms a real image. The magnitude of V_m is _____ $cm\ s^{-1}$.

[JEE (Advanced) 2022]



PGO105

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28. A plane polarized blue light ray is incident on a prism such that there is no reflection from the surface of the prism. The angle of deviation of the emergent ray is $\delta = 60°$ (see Figure-1). The angle of minimum deviation for red light from the same prism is $\delta_{min} = 30°$ (see Figure–2). The refractive index of the prism material for blue light is $\sqrt{3}$. Which of the following statement(s) is(are) correct?

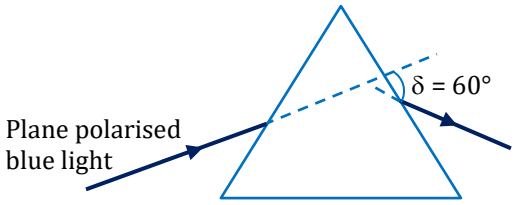


Figure-1

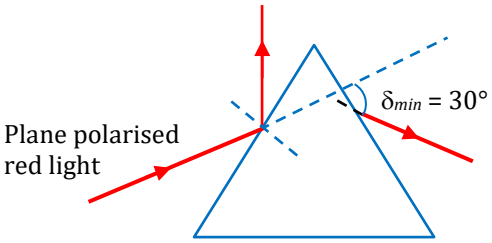


Figure-2

[JEE (Advanced) 2023]

(A) The blue light is polarized in the plane of incidence.

(B) The angle of the prism is 45°.

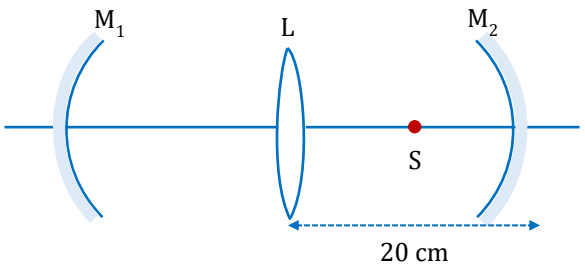
(C) The refractive index of the material of the prism for red light is $\sqrt{2}$.

(D) The angle of refraction for blue light in air at the exit plane of the prism is 60°.

PGO262

29. An optical arrangement consists of two concave mirrors M_1 and M_2, and a convex lens L with a common principal axis, as shown in the figure. The focal length of L is 10 *cm*. The radii of curvature of M_1 and M_2 are 20 *cm* and 24 *cm*, respectively. The distance between L and M_2 is 20 *cm*. A point object S is placed at the mid-point between L and M_2 on the axis. When the distance between L and M_1 is $n/7$ *cm*, one of the images coincides with S. The value of n is _____.

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[JEE (Advanced) 2023]

PGO263

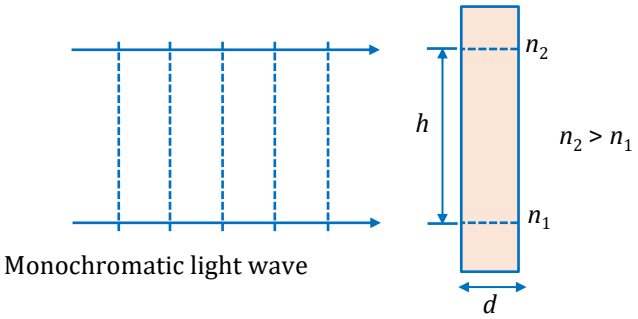
30. In an experiment for determination of the focal length of a thin convex lens, the distance of the object from the lens is 10 ± 0.1 *cm* and the distance of its real image from the lens is 20 ± 0.2 *cm*. The error in the determination of focal length of the lens is n %. The value of n is _____.

[JEE (Advanced) 2023]

PGO264

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31. A monochromatic light wave is incident normally on a glass slab of thickness d, as shown in the figure. The refractive index of the slab increases linearly from n_1 to n_2 over the height h. Which of the following statement(s) is (are) true about the light wave emerging out of the slab ?



(A) It will deflect up by an angle $tan^{-1}\left[\frac{(n_2^2-n_1^2)d}{2h}\right]$

(B) It will deflect up by an angle $tan^{-1}\left[\frac{(n_2-n_1)d}{h}\right]$

(C) It will not deflect.

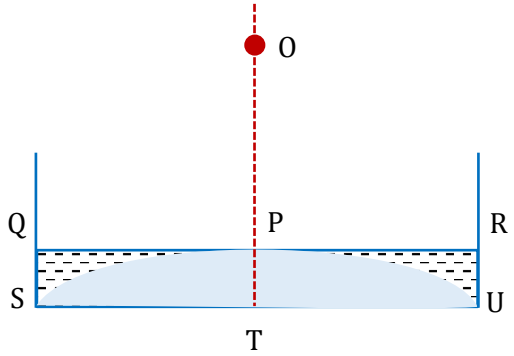
(D) The deflection angle depends only on $(n_2 - n_1)$ and not on the individual values of n_1 and n_2.

[JEE (Advanced) 2023]

PGO265

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32. A glass beaker has a solid, plano-convex base of refractive index 1.60, as shown in the figure. The radius of curvature of the convex surface (*SPU*) is 9 *cm*, while the planar surface (*STU*) acts as a mirror. This beaker is filled with a liquid of refractive index n up to the level *QPR*. If the image of a point object *O* at a height of h (*OT* in the figure) is formed onto itself, then, which of the following option(s) is(are) correct? **[JEE (Advanced) 2024]**



(A) For $n = 1.42$, $h = 50$ *cm*

(B) For $n = 1.35$, $h = 36$ *cm*

(C) For $n = 1.45$, $h = 65$ *cm*

(D) For $n = 1.48$, $h = 85$ *cm*

PGO273

For More Join: @IITJEE_Advanced

33. A light ray is incident on the surface of a sphere of refractive index n at an angle of incidence θ_0. The ray partially refracts into the sphere with angle of refraction ϕ_0 and then partly reflects from the back surface. The reflected ray then emerges out of the sphere after a partial refraction. The total angle of deviation of the emergent ray with respect to the incident ray is α. Match the quantities mentioned in List-I with their values in List-II and choose the correct option.

[JEE (Advanced) 2024]

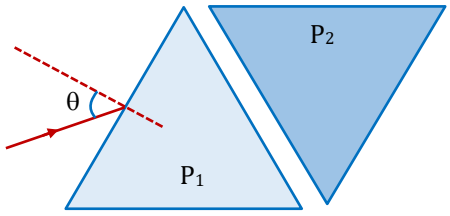
	List-I		**List-II**
(P)	If $n = 2$ and $\alpha = 180°$, then all the possible values of θ_0 will be	(1)	30° and 0°
(Q)	If $n = \sqrt{3}$ and $\alpha = 180°$, then all the possible values of θ_0 will be	(2)	60° and 0°
(R)	If $n = \sqrt{3}$ and $\alpha = 180°$, then all the possible values of ϕ_0 will be	(3)	45° and 0°
(S)	If $n = \sqrt{2}$ and $\theta_0 = 45°$, then all the possible values of α will be	(4)	150°
		(5)	0°

(A) P → 5; Q → 2; R → 1; S → 4
(B) P → 5; Q → 1; R → 2; S → 4
(C) P → 3; Q → 2; R → 1; S → 4
(D) P → 3; Q → 1; R → 2; S → 5

PGO274

34. Two equilateral-triangular prisms P_1 and P_2 are kept with their sides parallel to each other, in vacuum, as shown in the figure. A light ray enters prism P_1 at an angle of incidence θ such that the outgoing ray undergoes minimum deviation in prism P_2. If the respective refractive indices of P_1 and P_2 are $\sqrt{\frac{3}{2}}$ and $\sqrt{3}$, then $\theta = \sin^{-1}\left[\sqrt{\frac{3}{2}}\sin\left(\frac{\pi}{\beta}\right)\right]$, where the value of β is _____.

[JEE (Advanced) 2024]



PGO275

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	B	B	A	C	A	C	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	D	C	C	A	C	C	B	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	D	A	C	B	B	A	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	B	C	C	D	A	A	B	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	B	B	C	C	D	A	B	B	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABCD	BD	ABD	AD	BCD	AC	AC	BD	BC	AB
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	AB	AD	AC	BD	AC	ABCD	A	B	A	B
Que.	21	22	23	24			25			
Ans.	A	C	D	(A-Q), (B-P), (C-S), (D-R)			(A-P), (B-Q), (C-RT), (D-PT)			

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EXERCISE (S)

1. 3.75 **2.** 9 **3.** 7.00 **4.** 4

5. 25 **6.** 16 mm^3 **7.** 5 **8.** 2 mm

9. 8 **10.** $\theta < \sin^{-1}(2\sin 15°)$ **11.** (a) $\sin^{-1}\left(\frac{1}{5}\right)$ (b) air

12. $\frac{8}{3}$ mm **13.** 4 **14.** 14.00 **15.** 7.50

16. (i) 0.2 m, (ii) 0.4 m **17.** 17.5 cm **18.** 30

19. $\pi/4$ cm^2 **20.** 1 **21.** 4

22. (a) $f = -20$ cm, (b) + 80 cm, convergent achromatic lens **23.** 24.00

24. 14.60 **25.** 101 **26.** 25

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	4	3	4.48	476	1	2	1	50
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	12	10	32	2	4	1	31	2	3

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EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	7	B	2	AC	D	A	ABD	AD	A	ACD
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	8	130 [129.95, 130.05]	D	ACD	50	CD	1.50	AD	2	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	BCD	BC	6	AB	A	4	3	ACD	80 or 150 or 220	1
Que.	31	32	33	34						
Ans.	BD	AB	A	12						

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